

ALFAAS60220

## tert-Butyllithium, 1.3 M solution in pentane

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

产品说明: 叔丁基锂, 1.3 M正戊烷溶液  
 Product Description: tert-Butyllithium, 1.3 M solution in pentane

Cat No. : S60220

Supplier Avocado Research Chemicals Ltd.  
 (Part of Thermo Fisher Scientific)  
 Shore Road, Heysham  
 Lancashire, LA3 2XY,  
 United Kingdom  
 Office Tel: +44 (0) 1524 850506  
 Office Fax: +44 (0) 1524 850608

Emergency Telephone Number For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
 Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

E-mail address begel.sdsdesk@thermofisher.com

Recommended Use Laboratory chemicals.  
 Uses advised against No Information available

### SECTION 2. HAZARD IDENTIFICATION

**Physical State**  
 Liquid

**Appearance**  
 No information available

**Odor**  
 No information available

#### Emergency Overview

Highly flammable liquid and vapor. Catches fire spontaneously if exposed to air. In contact with water releases flammable gases which may ignite spontaneously. May be fatal if swallowed and enters airways. May cause drowsiness and dizziness. Extremely flammable liquid and vapor. May be harmful in contact with skin. Causes severe skin burns and eye damage. Toxic to aquatic life with long lasting effects. Reacts violently with water. Moisture sensitive. Air sensitive.

#### Classification of the substance or mixture

|                                                                        |                       |
|------------------------------------------------------------------------|-----------------------|
| Flammable liquids.                                                     | Category 2 Category 1 |
| Substances/mixtures which, in contact with water, emit flammable gases | Category 1            |
| Pyrophoric liquids                                                     | Category 1            |
| Aspiration Toxicity                                                    | Category 1            |
| Acute Dermal Toxicity                                                  | Category 5            |
| Skin Corrosion/Irritation                                              | Category 1 B          |
| Serious Eye Damage/Eye Irritation                                      | Category 1            |
| Specific target organ toxicity - (single exposure)                     | Category 3            |
| Acute aquatic toxicity                                                 | Category 2            |
| Chronic aquatic toxicity                                               | Category 2            |

#### Label Elements

# SAFETY DATA SHEET

## tert-Butyllithium, 1.3 M solution in pentane

**Signal Word****Danger****Hazard Statements**

H225 - Highly flammable liquid and vapor  
H224 - Extremely flammable liquid and vapor  
H260 - In contact with water releases flammable gases which may ignite spontaneously  
H250 - Catches fire spontaneously if exposed to air  
H304 - May be fatal if swallowed and enters airways  
H336 - May cause drowsiness or dizziness  
H313 - May be harmful in contact with skin  
H314 - Causes severe skin burns and eye damage  
H411 - Toxic to aquatic life with long lasting effects

**Precautionary Statements****Prevention**

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
P222 - Do not allow contact with air  
P231 + P232 - Handle and store contents under inert gas. Protect from moisture  
P240 - Ground and bond container and receiving equipment  
P242 - Use non-sparking tools  
P243 - Take action to prevent static discharges  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P271 - Use only outdoors or in a well-ventilated area  
P280 - Wear protective gloves/protective clothing/eye protection/face protection

**Response**

P302 + P334 - IF ON SKIN: Immerse in cool water or wrap in wet bandages  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P310 - Immediately call a POISON CENTER or doctor  
P330 - Rinse mouth  
P331 - Do NOT induce vomiting  
P363 - Wash contaminated clothing before reuse  
P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish

**Storage**

P402 + P404 - Store in a dry place. Store in a closed container  
P422 - Store contents under inert gas

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Physical and Chemical Hazards**

Vapors may cause flash fire or explosion. Highly flammable. Catches fire spontaneously if exposed to air. Reacts violently with water, liberating extremely flammable gases. Extremely flammable. Reacts violently with water.

**Health Hazards**

Aspiration hazard if swallowed - can enter lungs and cause damage. May cause drowsiness or dizziness. May be harmful in contact with skin. Corrosive. Causes skin and eye burns.

**Environmental hazards**

Toxic to aquatic life with long lasting effects. Reacts violently with water.

**Other Hazards**

This product does not contain any known or suspected endocrine disruptors. Toxic to terrestrial vertebrates.

**SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS**

# SAFETY DATA SHEET

## tert-Butyllithium, 1.3 M solution in pentane

| Component                     | CAS No   | Weight % |
|-------------------------------|----------|----------|
| Pentane                       | 109-66-0 | 87       |
| Lithium, (1,1-dimethylethyl)- | 594-19-4 | 13       |

### SECTION 4. FIRST AID MEASURES

#### General Advice

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

#### Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required.

#### Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Call a physician immediately.

#### Inhalation

If not breathing, give artificial respiration. Remove from exposure, lie down. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Call a physician immediately. Risk of serious damage to the lungs (by aspiration).

#### Ingestion

Do NOT induce vomiting. Clean mouth with water. Never give anything by mouth to an unconscious person. Call a physician immediately. Call a physician or poison control center immediately. If vomiting occurs naturally, have victim lean forward.

#### Most important symptoms and effects

Causes burns by all exposure routes. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting: Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation

#### Self-Protection of the First Aider

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

#### Notes to Physician

Treat symptomatically. Symptoms may be delayed.

### SECTION 5. FIRE-FIGHTING MEASURES

#### Suitable Extinguishing Media

CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam. Water mist may be used to cool closed containers.

#### Extinguishing media which must not be used for safety reasons

Water.

#### Specific Hazards Arising from the Chemical

Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin and mucous membranes. Reacts violently with water. Extremely flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

#### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

### SECTION 6. ACCIDENTAL RELEASE MEASURES

# SAFETY DATA SHEET

## tert-Butyllithium, 1.3 M solution in pentane

### Personal Precautions

Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. Remove all sources of ignition. Take precautionary measures against static discharges.

### Environmental Precautions

Do not flush into surface water or sanitary sewer system. Should not be released into the environment. Do not allow material to contaminate ground water system.

### Methods for Containment and Clean Up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Do not expose spill to water. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7. HANDLING AND STORAGE

### Handling

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance. Do not allow contact with water. Keep away from open flames, hot surfaces and sources of ignition. Use spark-proof tools and explosion-proof equipment. Use only non-sparking tools. To avoid ignition of vapors by static electricity discharge, all metal parts of the equipment must be grounded. Take precautionary measures against static discharges.

### Storage

Keep refrigerated. Store under an inert atmosphere. Corrosives area. Keep away from water or moist air. Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from heat, sparks and flame.

### Specific Use(s)

Use in laboratories

## SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

| Component | China                                                      | Taiwan                                      | Thailand      | Hong Kong                                   |
|-----------|------------------------------------------------------------|---------------------------------------------|---------------|---------------------------------------------|
| Pentane   | TWA: 500 mg/m <sup>3</sup><br>STEL: 1000 mg/m <sup>3</sup> | TWA: 600 ppm<br>TWA: 1770 mg/m <sup>3</sup> | TWA: 1000 ppm | TWA: 600 ppm<br>TWA: 1770 mg/m <sup>3</sup> |

| Component | ACGIH TLV     | OSHA PEL                                                                                                                                                                             | NIOSH                                                                                                               | The United Kingdom                                                                                                    | European Union                                           |
|-----------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| Pentane   | TWA: 1000 ppm | (Vacated) TWA: 600 ppm<br>(Vacated) TWA: 1800 mg/m <sup>3</sup><br>(Vacated) STEL: 750 ppm<br>(Vacated) STEL: 2250 mg/m <sup>3</sup><br>TWA: 1000 ppm<br>TWA: 2950 mg/m <sup>3</sup> | IDLH: 1500 ppm<br>TWA: 120 ppm<br>TWA: 350 mg/m <sup>3</sup><br>Ceiling: 610 ppm<br>Ceiling: 1800 mg/m <sup>3</sup> | STEL: 1800 ppm 15 min<br>STEL: 5400 mg/m <sup>3</sup> 15 min<br>TWA: 600 ppm 8 hr<br>TWA: 1800 mg/m <sup>3</sup> 8 hr | TWA: 1000 ppm (8hr)<br>TWA: 3000 mg/m <sup>3</sup> (8hr) |

### Legend

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

### Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents. MDHS70 General methods for sampling airborne gases and vapours MDHS 88 Volatile organic compounds in air. Laboratory method using diffusive samplers, solvent desorption and gas chromatography MDHS

# SAFETY DATA SHEET

## tert-Butyllithium, 1.3 M solution in pentane

96 Volatile organic compounds in air - Laboratory method using pumped solid sorbent tubes, solvent desorption and gas chromatography MDHS 91 Metals and metalloids in workplace air by X-ray fluorescence spectrometry MDHS 99 Metals in air by ICP-AES

### Exposure Controls

#### Engineering Measures

Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas. Use explosion-proof electrical/ventilating/lighting equipment. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source.

#### Personal protective equipment

**Eye Protection** Goggles (European standard - EN 166)

**Hand Protection** Protective gloves

| Glove material | Breakthrough time                 | Glove thickness | EU standard | Glove comments        |
|----------------|-----------------------------------|-----------------|-------------|-----------------------|
| Viton (R)      | See manufacturers recommendations | -               | EN 374      | (minimum requirement) |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection** Long sleeved clothing

**Respiratory Protection** When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Large scale/emergency use** Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced  
**Recommended Filter type:** low boiling organic solvent Type AX Brown conforming to EN371 or Organic gases and vapours filter Type A Brown conforming to EN14387

**Small scale/Laboratory use** Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.  
**Recommended half mask:-** Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141  
 When RPE is used a face piece Fit Test should be conducted

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls** Prevent product from entering drains. Do not allow material to contaminate ground water system.

## SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

#### Appearance

**Physical State** Liquid

**Odor** No information available

**Odor Threshold** No data available

**pH** No information available

**Melting Point/Range** No data available

**Softening Point** No data available

# SAFETY DATA SHEET

## tert-Butyllithium, 1.3 M solution in pentane

|                                                |                          |                                             |
|------------------------------------------------|--------------------------|---------------------------------------------|
| <b>Boiling Point/Range</b>                     | No information available |                                             |
| <b>Flash Point</b>                             | -49 °C / -56.2 °F        | <b>Method</b> - No information available    |
| <b>Evaporation Rate</b>                        | No data available        |                                             |
| <b>Flammability (solid,gas)</b>                | Not applicable           | Liquid                                      |
| <b>Explosion Limits</b>                        | No data available        |                                             |
| <b>Vapor Pressure</b>                          | No data available        |                                             |
| <b>Vapor Density</b>                           | No data available        | (Air = 1.0)                                 |
| <b>Specific Gravity / Density</b>              | 0.652                    |                                             |
| <b>Bulk Density</b>                            | Not applicable           | Liquid                                      |
| <b>Water Solubility</b>                        | No information available |                                             |
| <b>Solubility in other solvents</b>            | No information available |                                             |
| <b>Partition Coefficient (n-octanol/water)</b> |                          |                                             |
| <b>Component</b>                               | <b>log Pow</b>           |                                             |
| Pentane                                        | 3.45                     |                                             |
| <b>Autoignition Temperature</b>                | No data available        |                                             |
| <b>Decomposition Temperature</b>               | No data available        |                                             |
| <b>Viscosity</b>                               | No data available        |                                             |
| <b>Explosive Properties</b>                    |                          | Vapors may form explosive mixtures with air |
| <b>Oxidizing Properties</b>                    | No information available |                                             |

### SECTION 10. STABILITY AND REACTIVITY

|                                         |                                                                                                                         |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>Stability</b>                        | Moisture sensitive. Air sensitive. May form precipitate.                                                                |
| <b>Hazardous Reactions</b>              | None under normal processing. Reacts violently with water.                                                              |
| <b>Hazardous Polymerization</b>         | No information available.                                                                                               |
| <b>Conditions to Avoid</b>              | Exposure to moist air or water. Exposure to moisture. Keep away from open flames, hot surfaces and sources of ignition. |
| <b>Materials to avoid</b>               | No information available.                                                                                               |
| <b>Hazardous Decomposition Products</b> | Carbon oxides. Lithium oxide.                                                                                           |

### SECTION 11. TOXICOLOGICAL INFORMATION

#### Product Information

(a) acute toxicity;  
Toxicology data for the components

| Component | LD50 Oral                 | LD50 Dermal                  | LC50 Inhalation                         |
|-----------|---------------------------|------------------------------|-----------------------------------------|
| Pentane   | LD50 > 2000 mg/kg ( Rat ) | LD50 = 3000 mg/kg ( Rabbit ) | LC50 = 364 g/m <sup>3</sup> ( Rat ) 4 h |

(b) skin corrosion/irritation; Category 1 B

(c) serious eye damage/irritation; Category 1

(d) respiratory or skin sensitization;  
Respiratory No data available  
Skin No data available

(e) germ cell mutagenicity; No data available

# SAFETY DATA SHEET

## tert-Butyllithium, 1.3 M solution in pentane

(f) carcinogenicity; No data available  
There are no known carcinogenic chemicals in this product

(g) reproductive toxicity; No data available

(h) STOT-single exposure; Category 3  
Results / Target organs Central nervous system (CNS)

(i) STOT-repeated exposure; No data available  
Target Organs None known.

(j) aspiration hazard; Category 1

**Symptoms / effects, both acute and delayed** Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting: Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation

## SECTION 12. ECOLOGICAL INFORMATION

**Ecotoxicity effects** The product contains following substances which are hazardous for the environment. Toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. May cause long-term adverse effects in the environment. Do not allow material to contaminate ground water system.

| Component | Freshwater Fish                                                                                                                               | Water Flea                             | Freshwater Algae | Microtox |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|------------------|----------|
| Pentane   | LC50: = 9.99 mg/L, 96h (Lepomis macrochirus)<br>LC50: = 11.59 mg/L, 96h (Pimephales promelas)<br>LC50: = 9.87 mg/L, 96h (Oncorhynchus mykiss) | EC50: = 9.74 mg/L, 48h (Daphnia magna) |                  |          |

**Persistence and Degradability** Product contains heavy metals. Discharge into the environment must be avoided. Special pre-treatment is necessary  
**Persistence** May persist, based on information available.  
**Degradation in sewage treatment plant** Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

**Bioaccumulative Potential** May have some potential to bioaccumulate

| Component | log Pow | Bioconcentration factor (BCF) |
|-----------|---------|-------------------------------|
| Pentane   | 3.45    | No data available             |

**Mobility in soil** No information available

**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors  
**Persistent Organic Pollutant** This product does not contain any known or suspected substance  
**Ozone Depletion Potential** This product does not contain any known or suspected substance

# SAFETY DATA SHEET

tert-Butyllithium, 1.3 M solution in pentane

## SECTION 13. DISPOSAL CONSIDERATIONS

|                                            |                                                                                                                                                                                                                                                                                                                                            |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste from Residues/Unused Products</b> | Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.                                                                                                                                                                     |
| <b>Contaminated Packaging</b>              | Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.                                                                                                   |
| <b>Other Information</b>                   | Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Can be landfilled or incinerated, when in compliance with local regulations. Do not empty into drains. Large amounts will affect pH and harm aquatic organisms. Do not let this chemical enter the environment. |

## SECTION 14. TRANSPORT INFORMATION

### Road and Rail Transport

|                                |                                                              |
|--------------------------------|--------------------------------------------------------------|
| <b>UN-No</b>                   | UN3394                                                       |
| <b>Proper Shipping Name</b>    | ORGANOMETALLIC SUBSTANCE, LIQUID, PYROPHORIC, WATER-REACTIVE |
| <b>Technical Shipping Name</b> | tert-Butyllithium/entane                                     |
| <b>Hazard Class</b>            | 4.2                                                          |
| <b>Subsidiary Hazard Class</b> | 4.3                                                          |
| <b>Packing Group</b>           | I                                                            |

### IMDG/IMO

|                                |                                                              |
|--------------------------------|--------------------------------------------------------------|
| <b>UN-No</b>                   | UN3394                                                       |
| <b>Proper Shipping Name</b>    | ORGANOMETALLIC SUBSTANCE, LIQUID, PYROPHORIC, WATER-REACTIVE |
| <b>Technical Shipping Name</b> | tert-Butyllithium/entane                                     |
| <b>Hazard Class</b>            | 4.2                                                          |
| <b>Subsidiary Hazard Class</b> | 4.3                                                          |
| <b>Packing Group</b>           | I                                                            |

### IATA

|                                |                          |
|--------------------------------|--------------------------|
| <b>Technical Shipping Name</b> | tert-Butyllithium/entane |
|--------------------------------|--------------------------|

|                                     |                                 |
|-------------------------------------|---------------------------------|
| <b>Special Precautions for User</b> | No special precautions required |
|-------------------------------------|---------------------------------|

## SECTION 15. REGULATORY INFORMATION

### International Inventories

China, X = listed, Australia, U.S.A. (TSCA), Canada (DSL/NDSL), Europe (EINECS/ELINCS/NLP), Australia (AICS), Korea (KECL), China (IECSC), Japan (ENCS), Philippines (PICCS), Taiwan (TCSI), Japan (ISHL), New Zealand (NZIoC), Japan (ISHL).

| Component                     | The Inventory of Hazardous Chemicals (2015 Edition) | List of dangerous goods GB 12268 - 2012 | TCSI | IECSC | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL        |
|-------------------------------|-----------------------------------------------------|-----------------------------------------|------|-------|-----------|------|-----|-------|------|------|------|-------------|
| Pentane                       | X                                                   | X                                       | X    | X     | 203-692-4 | X    | X   | X     | X    | X    | X    | KE-27968    |
| Lithium, (1,1-dimethylethyl)- | -                                                   | X                                       | X    | X     | 209-831-5 | X    | -   | X     | X    | X    | X    | 2014-3-6117 |

### National Regulations



# SAFETY DATA SHEET

## tert-Butyllithium, 1.3 M solution in pentane

### SECTION 16. OTHER INFORMATION

**Prepared By** Health, Safety and Environmental Department  
**Creation Date** 19-Apr-2023  
**Revision Date** 13-May-2024  
**Revision Summary** New emergency telephone response service provider.

#### Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Chemical incident response training.

Fire prevention and fighting, identifying hazards and risks, static electricity, explosive atmospheres posed by vapours and dusts.

#### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**PNEC** - Predicted No Effect Concentration

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

#### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadviser - LOLI, Merck index, RTECS

**Physical hazards** On basis of test data  
**Health Hazards** Calculation method  
**Environmental hazards** Calculation method

#### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**SAFETY DATA SHEET**  
**tert-Butyllithium, 1.3 M solution in pentane**

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**End of Safety Data Sheet**